Cooling cravers to the carefree: Behavioral archetypes of thermal preference from city-scale longitudinal smartwatch micro-surveys

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Abstract

Thermal comfort in buildings has long been described through physiological and heat-balance models, yet how people actually seek, avoid, and use cooling varies more than those models predict, especially in warm climates where air conditioning is pervasive. This paper asks whether individuals can be grouped by their thermal-preference behavior, using a longitudinal smartwatch micro-survey deployment in Singapore built on the open-source Cozie platform. The analysis draws on 9,358 geolocated thermal-preference responses from 104 eligible participants, each paired with a matched outdoor heat index and an environmental-control context based on how likely the occupied space is to be air-conditioned. Three person-level features (the rate of preferring cooler conditions, the rate of preferring warmer conditions, and the sensitivity of cooling demand to outdoor heat) are stabilized with empirical-Bayes shrinkage and clustered into four behavioral archetypes: the *Craver*, the *Captive*, the *Conventional*, and the *Carefree*. Cluster-bootstrap analysis shows that only the *Conventional* group's cooling demand rises with outdoor heat, although every group except the *Craver* asks for cooling less often in spaces likely to be air-conditioned. The *Craver* keeps asking for cooling whatever the heat or setting, the *Carefree* seldom do even outdoors in above-median heat, and the *Captive* report being too cold far more often than anyone else. The groups are stable under clustering perturbations and are not explained by wearable physiology or by most stable personal traits; only personality openness shows a moderate association. Thermal preference in this warm climate therefore appears to be only partly a physiological response, which argues for comfort strategies and cooling policies that let people choose among low-carbon ways of staying comfortable.

Keywords: thermal comfort, thermal preference, personal comfort, wearable sensing, ecological momentary assessment, behavioural archetypes

1. Introduction

Characterization of thermal comfort in the built environment has long been dominated by the creation of thermal comfort models, which combine physiology and environmental science to describe when and how a person should feel thermally comfortable or uncomfortable [1]. Rooted in heat-balance theory, these models relate metabolic heat production, clothing insulation, air temperature, radiant conditions, humidity, and air movement to a predicted comfort state, drawing on how core and skin temperatures jointly drive thermal sensation [2], how circadian phase and the thermoneutral zone set the body's baseline [3, 4], and how radiant and solar exposure adds heat that simple air-temperature metrics miss [5, 6]. From these inputs, the models predict thermal sensation, satisfaction, and preference, and they remain the backbone of the comfort standards and mechanical systems designed around them [7]. Newer adaptive formulations extend this picture by recognizing that people stay comfortable across a wider range of conditions than heat-balance theory alone would predict, as behavioral adjustment, expectation, and prior thermal history reshape what feels acceptable [8, 9]. Even within the same space, individuals differ systematically in thermal sensitivity and response [10, 11]. To capture this variation directly, personal comfort models learn from an individual's own repeated feedback and physiological signals rather than assuming a population average [12]. Yet both approaches face practical limits. Generalized models predict individual thermal sensation correctly only about a third of the time [7], while personal models are difficult to scale because they depend on collecting sustained, labeled data from every occupant [12]. Efforts to connect these models to the broader context that surrounds them, including the psychological and non-physical influences that shape perception [13, 14], demographic and cultural differences in expectation and preference [15, 16], and the social conventions and economics that govern who controls and pays for indoor climate [17, 18], remain nascent, but they may explain much of why the models still fall short.

1.1. Thermal behavior varies between people under similar conditions

Figure 1 sorts a non-exhaustive set of influences on thermal experience in cities into two broad families. The left side gathers the physical, environmental, and physiological factors that classical models were built to handle, although their detail and

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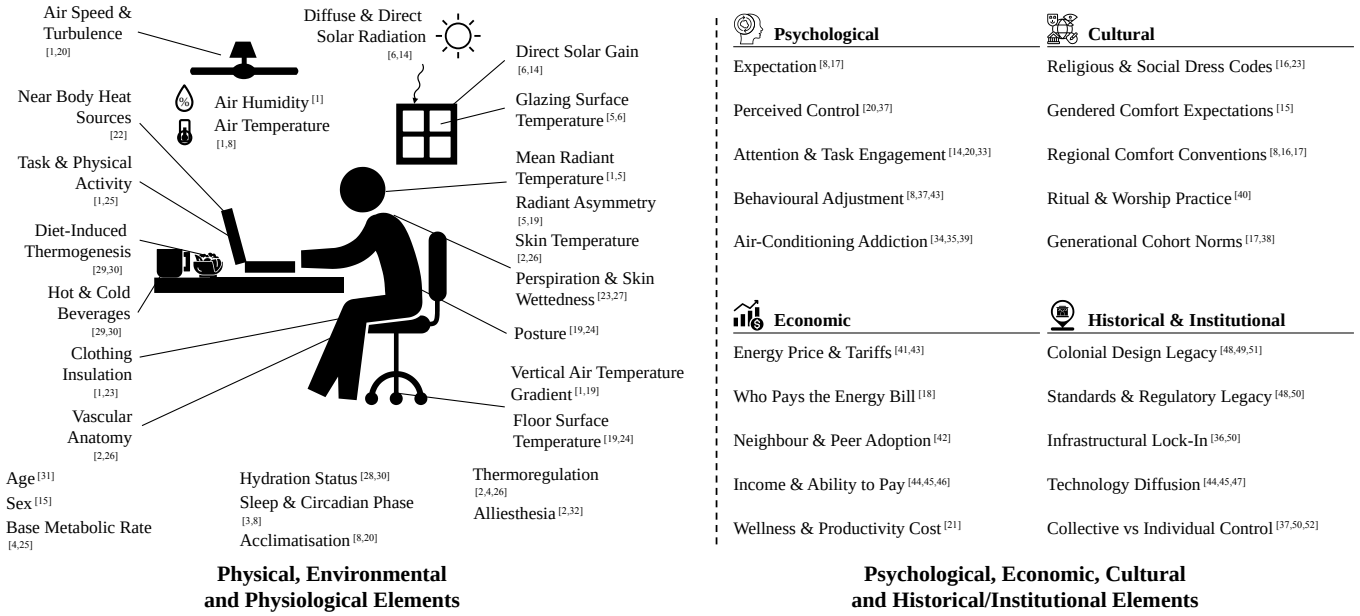


Figure 1: The many elements that shape individual thermal preference. Physical, environmental, and physiological influences are gathered on the left, with the psychological, cultural, economic, and historical or institutional influences that sit outside the heat balance on the right, together illustrating why identical conditions can feel different to different people.

interactions still complicate prediction. At the environmental scale, real spaces are rarely uniform, and local asymmetries in radiant, surface, and air-movement conditions shape comfort in ways a single room temperature cannot represent [19], while personally controlled air movement can keep occupants comfortable at warmer set points and partly offset the cognitive cost of heat [20], and office work performance in a simulated tropical office has been found to hold steady between 23 and 29 °C [21]. Closer to the body, near-body heat sources such as laptops [22], the insulation and cut of clothing [23], posture [24], and physical activity and metabolic rate [25] all shift the balance between heat production and loss. The physiological state of the occupant adds further variation through cutaneous blood flow [26], sweating and skin wettedness [27], hydration [28], the thermic effect of food and drink [29, 30], and age-related shifts in thermal thresholds [31]. Even the pleasantness of a given thermal stimulus depends on internal state, so the same temperature can feel welcome or aversive depending on whether it helps restore the body's balance [32]. These physical and physiological signals also rarely act alone, combining with the visual, acoustic, and air-quality exposures that any rigorous account of the indoor experience must weigh together [33].

The right side of Figure 1 gathers the psychological, cultural, economic, and historical or institutional forces that sit outside the heat balance yet strongly condition how people interpret and act on it. Psychologically, cooling choices are bound up with identity, values, and the situation at hand, not temperature alone [34], and habitual reliance on air conditioning can harden into a practice that people find difficult to step outside of, even briefly [35]. Culturally, expectations of what a comfortable indoor climate should feel like are learned and socially organized, shaped by evolving conventions, domestic technology, and the material culture of cooling [36, 37, 38], and they take on distinctive forms in specific settings, from the air-conditioned lives of young people in Singapore [39] to thermal comfort and cooling energy use in naturally and mechanically ventilated mosques [40]. Economically, the decision to buy and run cooling responds to weather, prices, incomes, and the efficiency of the available equipment [41, 42, 43], and as incomes rise and the climate warms, adoption is projected to grow steeply, with large and unequally distributed consequences for electricity demand and emissions [44, 45, 46, 47]. Behind these forces sits a historical and institutional legacy in which particular comfort norms, standards, and infrastructures were exported and standardized, homogenizing expectations across very different climates and cultures [48, 49, 50, 51], a trajectory whose sustainability in an increasingly hot world is now openly in question [52].

1.2. Repeated in-situ data can reveal behavioral response patterns

Capturing the convergence of these physical and non-physical influences is difficult in controlled or laboratory settings, where the very context that shapes perception is stripped away. Observing people in their natural environments over time is one of the few ways to see how thermal experience unfolds across the varied spaces, activities, and social situations of daily life. Repeated in-situ reporting through ecological momentary assessment provides a practical route to these observations, recording subjective states as they occur instead of in recollection [53]. The data analyzed here come from such a deployment in Singapore, which used the open-source Cozie smartwatch platform to deliver micro-surveys on an Apple

Watch and record the wearer's onboard physiological signals alongside each response [54]; each response was later linked to outdoor weather. The deployment grew out of a broader program that used smartwatch-based just-in-time adaptive interventions to nudge heat and noise mitigation in the urban environment [55, 56], and the resulting dataset has since supported a public machine-learning competition on predicting heat and noise perception from digital-twin context [57, 58] as well as related work linking wearable reports to urban digital twins [59, 60]. Previous research characterizes this same deployment of 9,962 geolocated micro-survey responses from 106 participants, showing that everyday urban spaces separate most sharply along behavioral and acoustic lines and that thermal preference tracks the immediate controlled environment more than the outdoor weather [61]. That space-level characterization motivates the present study, which expands the behavioral dimension of the same dataset by shifting the unit of analysis from the space to the person. Rather than asking how perception differs between spaces, this study clusters participants by their repeated thermal-preference behavior to uncover person-level archetypes, and then examines whether participant metadata offers clues to the non-physical and environmental elements of their thermal experience.

2. Methods

This study builds on a smartwatch micro-survey deployment developed and reported in several prior studies, which is summarized here before the archetype analysis that is new to this work. The data were collected in Singapore with the open-source Cozie platform, an Apple Watch application that administers short micro-surveys and records the wearer's onboard physiological signals [54], with each response subsequently matched to outdoor weather [61]. The deployment itself grew out of a smartwatch-driven just-in-time adaptive intervention framework for building occupants [55], and its instrumentation, recruitment, and preliminary results have been described in the methods and competition papers that accompanied it [57, 58]. Participants were recruited in and around the National University of Singapore under approval from its Institutional Review Board, gave informed consent, and wore a paired Apple Watch and iPhone during working hours for approximately four weeks while completing repeated micro-surveys [61]. The resulting privacy-protected dataset of 9,962 geolocated responses from 106 participants was released publicly for a machine-learning competition [57, 58] and is characterized at the level of urban space in the companion paper [61]. Within this dataset, the 9,520 responses that carry both a thermal-preference answer and a matched outdoor heat index form the basis for the present analysis, with a median of 96 responses per participant across the cohort. Reported preferences skew toward no change, cooler preferences are far more common than warmer ones, the home accounts for the largest share of responses (45%), and the matched outdoor heat index is high and narrow around a median near 34 °C, consistent with the warm tropical setting and characterized more fully in the companion paper [61]. Here the same underlying deployment is reused but analyzed at the level of the person, and the subsections below describe the survey measures and environmental-control context, the participant eligibility screen and person-level features, the archetype-discovery procedure, and the robustness checks used to scrutinize the resulting groupings.

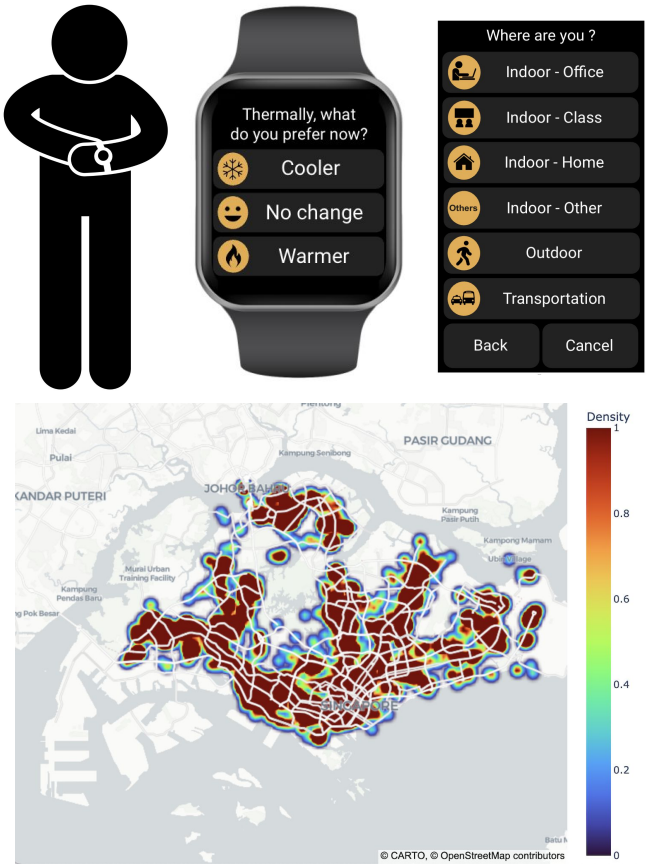


Figure 2: The Cozie smartwatch micro-survey interface, which prompts participants to report their thermal preference and current space type directly from the wrist (adapted from [61]).

2.1. *Survey measures, outdoor heat, and environmental-control context*

At each submission, the Cozie micro-survey presents a short battery of questions that records the participant's thermal preference and the level of nearby noise, together with conditional follow-ups on the kind of noise and earphone use, the location or space type, whether the participant is *Alone* or in a *Group*, and the activity being performed [61]. Figure 2 shows the two survey questions that this analysis uses as its primary components, thermal preference and location. The remainder of the battery, along with the linked physiological, weather, and urban-context data, is treated as outside the scope of this study. The thermal-preference question asks whether a participant would prefer to be *Cooler*, feel *No change*, or prefer to be *Warmer*, and it is answered on the watch alongside the

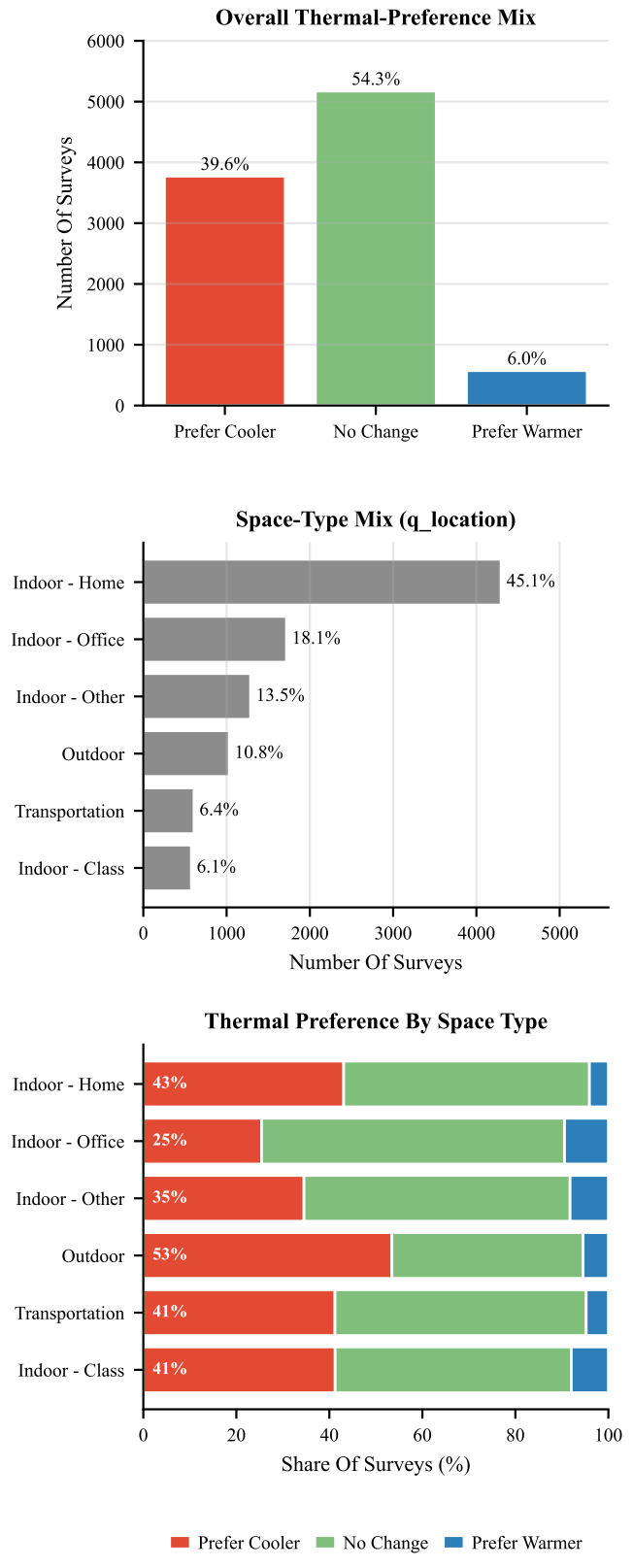


Figure 3: How thermal preference is distributed across the deployment. The panels trace the overall response mix (top), where surveys were collected across space types (middle), and how the preference mix shifts within each space type (bottom).

companion question that records the category of space they occupy at that moment. Thermal preference is the primary behavioral outcome throughout this study because it captures the desired direction of change rather than a sensation label, which maps more directly onto the adaptive actions a person might take. The location question resolves each response to an everyday space type, such as home, office, classroom, transport, and outdoor settings, so that preferences can be read in the context where they were expressed. Figure 3 summarizes the resulting survey composition, showing the overall balance of the three preference responses, how many surveys come from each space type, and how the preference mix shifts from one setting to another. To situate each response in its thermal context, each is joined to a one-hour mean outdoor heat index derived from matched local weather, which serves as an exposure proxy for the ambient conditions around the participant rather than a direct measurement of their individual microclimate. Because indoor spaces differ in how tightly their climate is controlled, each response is also assigned to an a-priori environmental-control context based on the likelihood that the occupied space is air-conditioned: outdoor settings are treated as having no air conditioning, home as possibly air-conditioned, offices, classrooms, and other indoor spaces as likely air-conditioned, and transport as very likely air-conditioned. This grouping makes it possible to ask whether a preference was expressed in a setting where the occupant could plausibly act on it through mechanical cooling, and it provides the control-context axis used in the mechanistic analysis that follows.

2.2. Participant eligibility and person-level features

Clustering participants requires a stable behavioral signal from each person, so the survey-level dataset is first restricted to participants who contributed enough responses on both sides of the heat range. Every response is labeled hot-side or cool-side relative to the cohort median of the one-hour outdoor heat index, which falls at 34.1 °C, so that each participant can be summarized by how their preferences differ between warmer and cooler exposures. A participant qualifies for the analysis if they submitted at least 15 responses in total, with at least 5 on the hot side and at least 5 on the cool side, applied across all locations rather than the home alone. This all-location screen retains 104 of the 106 participants (98%) and 9,358 of their responses, dropping only the two participants with too few hot-side responses.

For each qualifying participant, three person-level features are constructed that summarize their thermal-preference behavior. The first two are the rates at which they prefer to be cooler and prefer to be warmer, computed across all of their responses and stabilized with empirical-Bayes Beta–Binomial shrinkage, which fits a cohort-wide Beta prior by marginal maximum likelihood and pulls each participant's rate toward that prior in proportion to how few responses they contributed. This shrinkage prevents participants with sparse records from producing extreme rates that would otherwise dominate the clustering. Figure 4 shows the fitted cohort priors and how the shrinkage pulls each participant's raw rates toward those priors. The third feature is a within-participant heat sensitivity, defined as the dif-

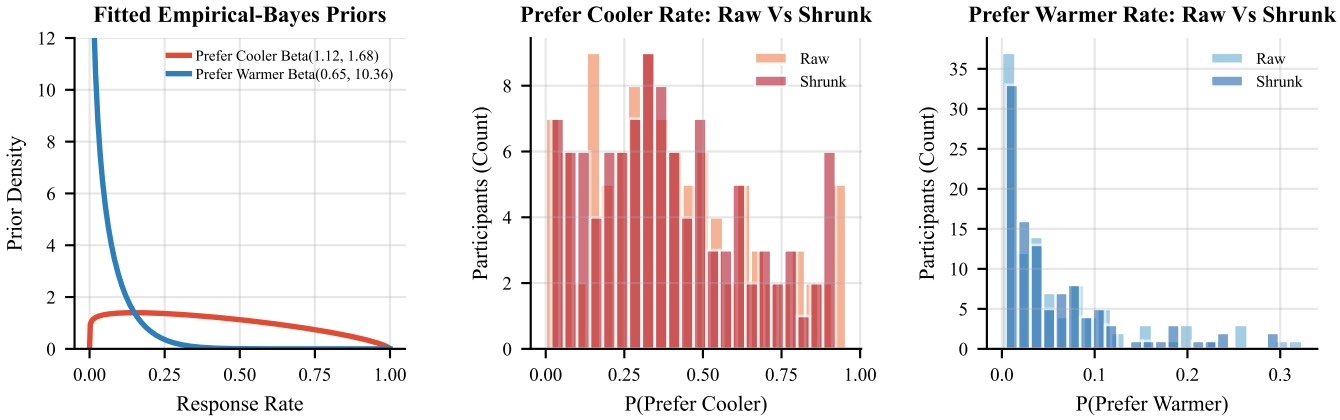


Figure 4: Constructing stable person-level features through empirical-Bayes shrinkage. Fitted Beta priors are shown on the left, and the center and right panels show how participants' raw prefer-cooler and prefer-warmer rates are pulled toward the cohort mean, most strongly for those with few responses.

ference between their prefer-cooler rate on the hot side and on the cool side, $P(\text{Cooler} \mid \text{hot}) - P(\text{Cooler} \mid \text{cool})$, and left unshrunk so that it reflects each person's raw responsiveness to outdoor heat. Together these features place each participant in a compact behavioral space of how much they seek cooling, how much they seek warmth, and how strongly their cooling demand tracks outdoor heat, which the next subsection uses to discover the archetypes.

2.3. Archetype discovery, interpretation and scrutiny analysis

With each qualifying participant reduced to the three behavioral features, the features are standardized to zero mean and unit variance so that no single rate dominates the distance metric, and participants are grouped with k-means clustering. To choose the number of archetypes without over-fitting, candidate solutions from two to six clusters are swept and evaluated by their mean silhouette width, using many restarts and a fixed random seed for determinism. The number of archetypes to carry forward is selected by balancing silhouette cohesion against interpretability, rather than treating the silhouette optimum as definitive. The resulting groups are then interpreted through their standardized feature centroids and a principal-component projection of the feature space, alongside per-cluster silhouette widths that indicate how internally coherent each group is. Because the underlying features vary continuously, the groups are treated as descriptive behavioral summaries, not categorical types, and membership is expected to be soft near the boundaries between them. The number of archetypes carried forward, their standardized centroids, and their descriptive names are presented in the Results.

Each archetype is then examined in its thermal and environmental context to test whether the group behaves as its feature profile implies. The dose-response of thermal preference to outdoor heat is examined by binning responses across the observed heat range, and preferences are compared across the a-priori environmental-control contexts, focusing on the gradient between likely air-conditioned and outdoor settings. These contrasts are quantified with distribution-free effect sizes, principally Cliff's delta, and uncertainty is attached with participant-level bootstrap confidence intervals rather than relying on p-values. The stability of the solution is assessed by re-clustering across random seeds and subsamples and by measuring how far membership agrees under alternative outdoor-heat thresholds. Because survey observations are nested within participants and were not experimentally assigned, every comparison is treated as descriptive and associational rather than causal.

3. Results

Applying the person-level feature construction and clustering to the 104 eligible participants yields four behavioral thermal-preference archetypes. The four groups, named the *Craver*, the *Captive*, the *Conventional*, and the *Carefree*, differ in how often they seek cooling, how often they report being too cold, and how strongly their preferences track outdoor heat. Figure 5 summarizes how the four archetypes differ along the three behavioral features. The *Craver*, or Constant Cool-Seekers (n = 23, 22.1%), prefers cooler conditions in roughly 74% of responses and carries a below-average heat sensitivity, seeking cooling whether or not outdoor heat is high. The *Captive*, or Often Over-Cooled (n = 12, 11.5%), stands apart through a much higher prefer-warmer rate, near 21% against only a few percent in the other groups, indicating that this group reports being too cold far more often than any other. The *Conventional*, or Normatively Responsive (n = 35, 33.7%), the largest group, shows the strongest heat sensitivity, so its prefer-cooler response rises and falls with outdoor heat, as thermoregulation would predict. The *Carefree*, or Heat-Unbothered (n = 34, 32.7%), sits below the cohort mean on all three features and reports no change in roughly 77% of responses, remaining comfortable even outdoors at comparable heat exposure.

Clustering the standardized prefer-cooler rate, prefer-warmer rate, and heat sensitivity with k-means gave the mean silhouette width highest at three clusters (0.322) and nearly identical

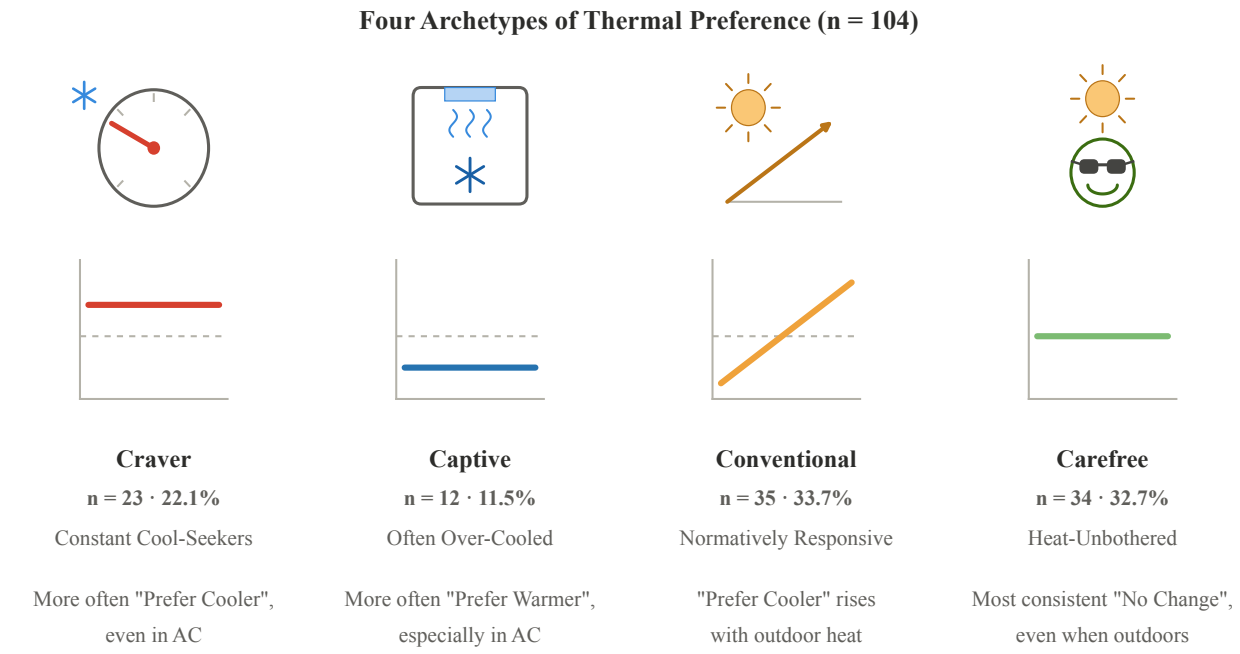


Figure 5: The four behavioral thermal-preference archetypes that emerge from participants' repeated feedback (n = 104): the *Craver* (Constant Cool-Seekers; n = 23, 22.1%), who keeps reaching for cooler even indoors; the *Captive* (Often Over-Cooled; n = 12, 11.5%), who is frequently left too cold; the *Conventional* (Normatively Responsive; n = 35, 33.7%), who tracks the heat as expected; and the *Carefree* (Heat-Unbothered; n = 34, 32.7%), who is largely content across conditions.

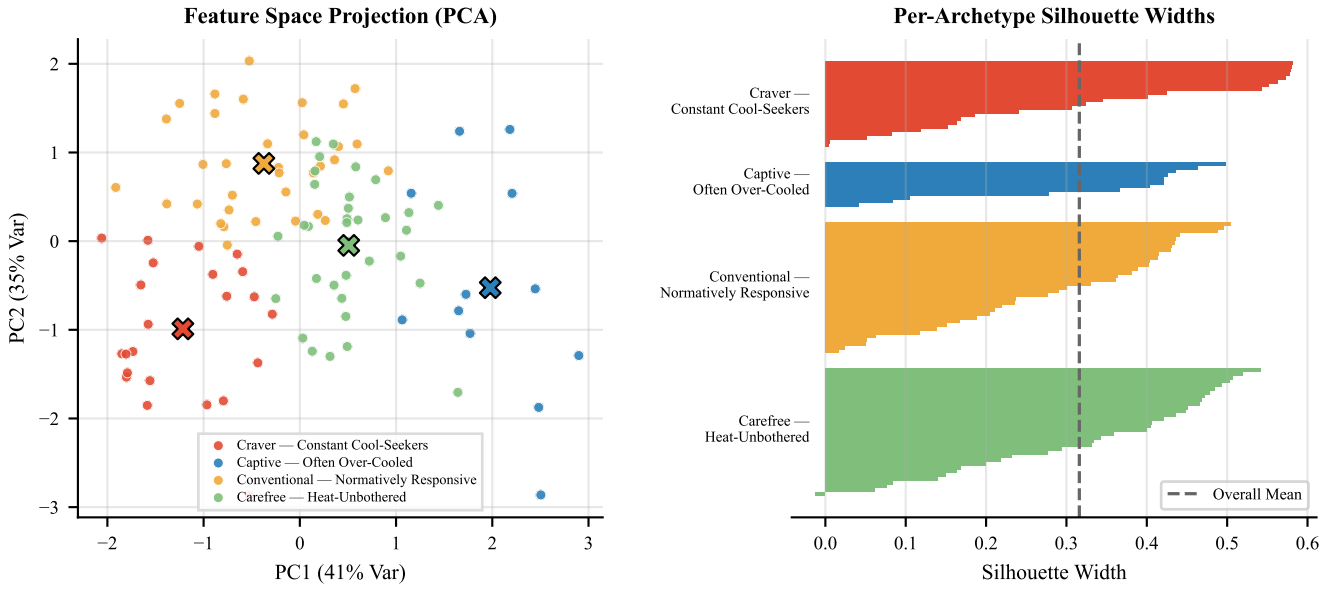


Figure 6: Separation diagnostics for the four archetypes: a principal-component projection of the standardized feature space (left) and participant silhouette widths by archetype (right).

at four (0.316). The four-cluster solution is carried forward because it separates an interpretively distinct group that the three-cluster solution merges, at negligible cost to cohesion, and it is not treated as an objectively optimal partition. Figure 6 shows that the four groups occupy distinct regions of the principal-component projection, with positive mean silhouette widths in every group (0.29 to 0.33), which indicates moderate rather than sharp separation. Figure 7 shows the standardized centroids of the four archetypes, with each group distinguished by a different feature: the *Craver* stands out on prefer-cooler rate,

the *Captive* on prefer-warmer rate, and the *Conventional* on heat sensitivity, while the *Carefree* sits below the cohort mean on all three.

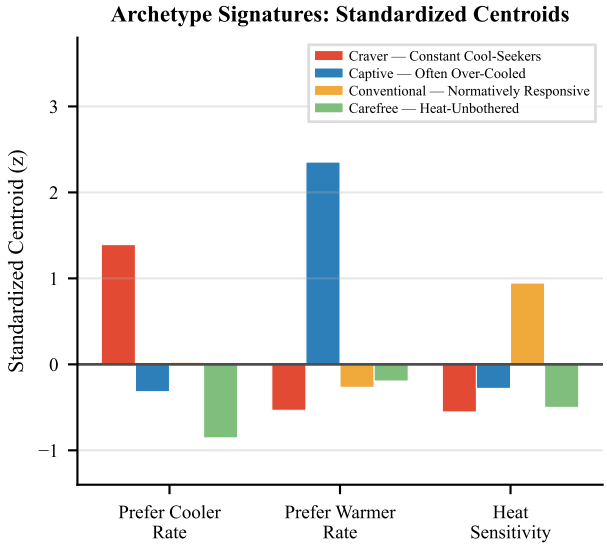


Figure 7: The behavioral signature of each archetype, drawn as standardized cluster centroids for prefer-cooler rate, prefer-warmer rate, and heat sensitivity.

3.1. Heat dose-response and environmental-control context

The four archetypes were derived from behavior alone, so this section places their responses back in thermal and environmental context to check whether each group behaves as its feature profile implies. It covers, in turn, how far each group's cooling demand falls as spaces become more likely to be air-conditioned, where over-cooling occurs, how preference responds to outdoor heat, and whether heat tolerance reflects lower heat exposure. Unless a caption states otherwise, uncertainty is reported as 95% cluster-bootstrap confidence intervals that resample whole participants, and group contrasts are summarized with distribution-free effect sizes, so that participant-nested responses are not treated as independent. Because participants were not assigned to contexts or heat levels, every result is read as descriptive and associational rather than causal.

Figure 8 shows the share of prefer-cooler responses for each archetype across five settings ordered by their air-conditioning likelihood, from outdoors with no air conditioning to offices, classrooms, and transport that are likely air-conditioned. For three of the four groups the prefer-cooler share falls as spaces become more likely to be air-conditioned, consistent with relief from mechanical cooling: the *Conventional* drops from about 62% outdoors to 33% in offices and classrooms, the *Captive* from 53% to 23%, and the *Carefree*, already the most heat-tolerant, from 32% to 11%. The *Craver* is the exception, holding a high prefer-cooler share of roughly 64 to 82% across every context, including the spaces most likely to be air-conditioned, which points to a cooling demand that mechanical conditioning does not satisfy. The confidence intervals are wider for the smaller groups, particularly the *Captive*, but the separation of the *Craver* above the other archetypes in the likely air-conditioned settings is consistent.

Figure 9 isolates the too-cold response by plotting the prefer-warmer share of each archetype across the control contexts. The *Captive* is the only group that reports being too cold with any regularity, preferring warmer conditions in about 14 to 28% of responses depending on the setting, against roughly 1 to 7% for every other archetype. Its over-cooling is most pronounced in the likely air-conditioned offices, classrooms, and other indoor spaces, where the prefer-warmer share reaches 28%, and lowest in transport, at about 14%. The within-*Captive* gradient from outdoors to these likely air-conditioned settings is positive, at about 7 percentage points, but its 95% cluster-bootstrap interval is wide and includes zero, so the level of over-cooling is clear while its context gradient remains uncertain in this small group of twelve participants.

Figure 10 traces the prefer-cooler share against outdoor heat-index sextiles within three control supergroups. Only the *Conventional* shows a clear upward slope, as expected for a group whose comfort tracks the weather. The slope is steepest outdoors, where no mechanical cooling intervenes and the *Conventional* prefer-cooler share climbs from under 30% in the coolest sextile to nearly 90% in the hottest, remains clear in home and other indoor settings, at about 21% to 58%, and largely flattens in likely air-conditioned settings. The *Craver* stays high and nearly flat and the *Carefree* stays low and nearly flat. The *Captive* curves are noisy rather than flat, rising outdoors, where each bin holds only 15 to 29 responses, and falling in likely air-conditioned settings, yet the group's overall heat coupling is indistinguishable from zero (Figure 11).

Figure 11 reduces the dose-response to a single heat-coupling value per archetype, the difference in prefer-cooler probability between responses above and below the cohort median heat index. Only the *Conventional* has a coupling distinguishable from zero, at about +0.18 (95% cluster-bootstrap interval 0.11 to 0.25), while the *Craver*, *Captive*, and *Carefree* sit near zero with intervals that span it, so only one archetype's cooling demand moves measurably with outdoor heat.

Figure 12 asks whether the Heat-Unbothered *Carefree* avoid heat or tolerate it. The left panel shows that outdoor-exposure share is statistically indistinguishable across the four archetypes, near 10 to 11% for each (Kruskal-Wallis $p = 0.88$), so no group is systematically sheltered from the heat. The right panel shows that even outdoors, and even when outdoors in above-median heat, the *Carefree* report no change far more often, in about 62% of outdoor responses and 54% of hot outdoor responses, against 21% and 21% for the *Craver*, 26% and 14% for the *Captive*, and 38% and 21% for the *Conventional*. Given their similar exposure, this points to heat tolerance in the *Carefree* rather than avoidance of hot conditions.

3.2. Robustness and boundary membership

Several checks test whether the four archetypes reflect real behavioral differences or merely the clustering procedure. The archetypes are strongly separated on the very features that define them, with pairwise Cliff's delta on the prefer-cooler

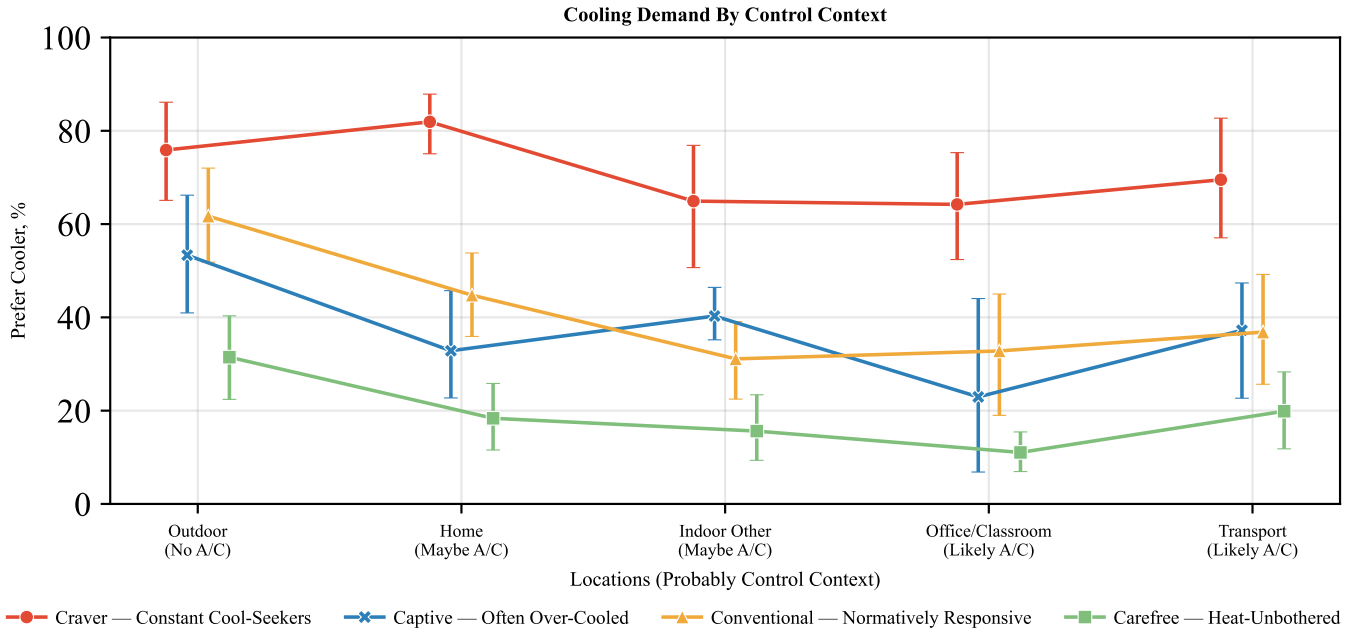


Figure 8: Whether air conditioning brings relief, shown as prefer-cooler responses by archetype across probable control contexts ordered by increasing air-conditioning likelihood, with 95 percent cluster-bootstrap confidence intervals.

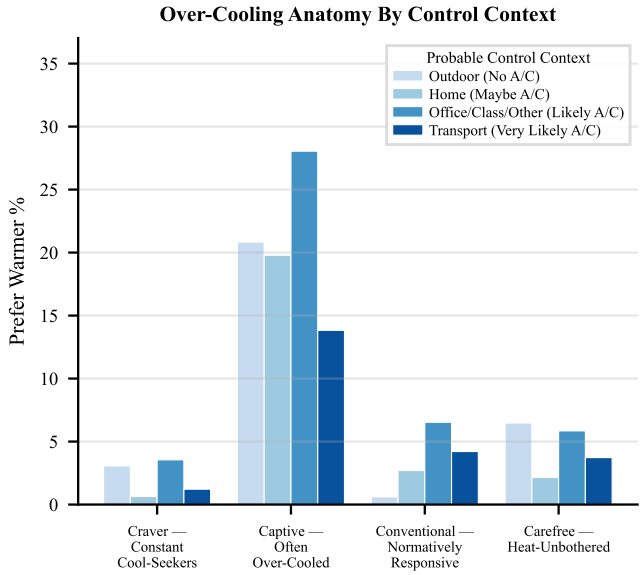


Figure 9: The over-cooling signal, shown as prefer-warmer responses by archetype across probable control contexts ordered by increasing air-conditioning likelihood.

rate approaching its extremes of ±1 between the most distinct groups. Figure 13 shows that the archetypes also vary in survey characteristics that were never used in the clustering, namely the mix of space types, noise distraction, and alone-versus-group context, which suggests the groups capture behavior beyond the three clustering features.

By contrast, Figure 14 shows that the archetypes are undifferentiated on wearable physiology in the hour before each survey, with Kruskal-Wallis effect sizes at or below 0.024 for the resting heart-rate proxy, mean heart rate, heart-rate standard deviation, and movement, and none of the differences statistically significant. This null result is consistent with thermal preference being a perceptual and behavioral signal rather than a simple function of heart rate or activity.

Figure 15 shows that among stable onboarding traits, only Ten-Item Personality Inventory Openness separates the archetypes at a medium effect size (epsilon-squared 0.137, $p = 0.003$), running highest in the *Carefree*, while the other personality dimensions, life satisfaction, sensory sensitivity, age, body mass index, years in Singapore, acclimatization, and gender all overlap across groups.

The partition itself is stable to reasonable perturbations, reproducing across fifty random seeds with a median adjusted Rand index of 0.94 and under repeated ninety-percent subsampling with a median adjusted Rand index of 0.77. It is more sensitive to the choice of heat threshold, with adjusted Rand indices between about 0.47 and 0.65 when the hot-cool split is moved from the cohort median to the 45th, 55th, or 60th percentile of heat index (computed over participants who qualify under both thresholds), which is expected because that threshold enters the feature construction itself. These results support reading the archetypes as a useful discretization of a continuous behavioral space rather than a deterministic classifier that would assign every new individual to a fixed type.

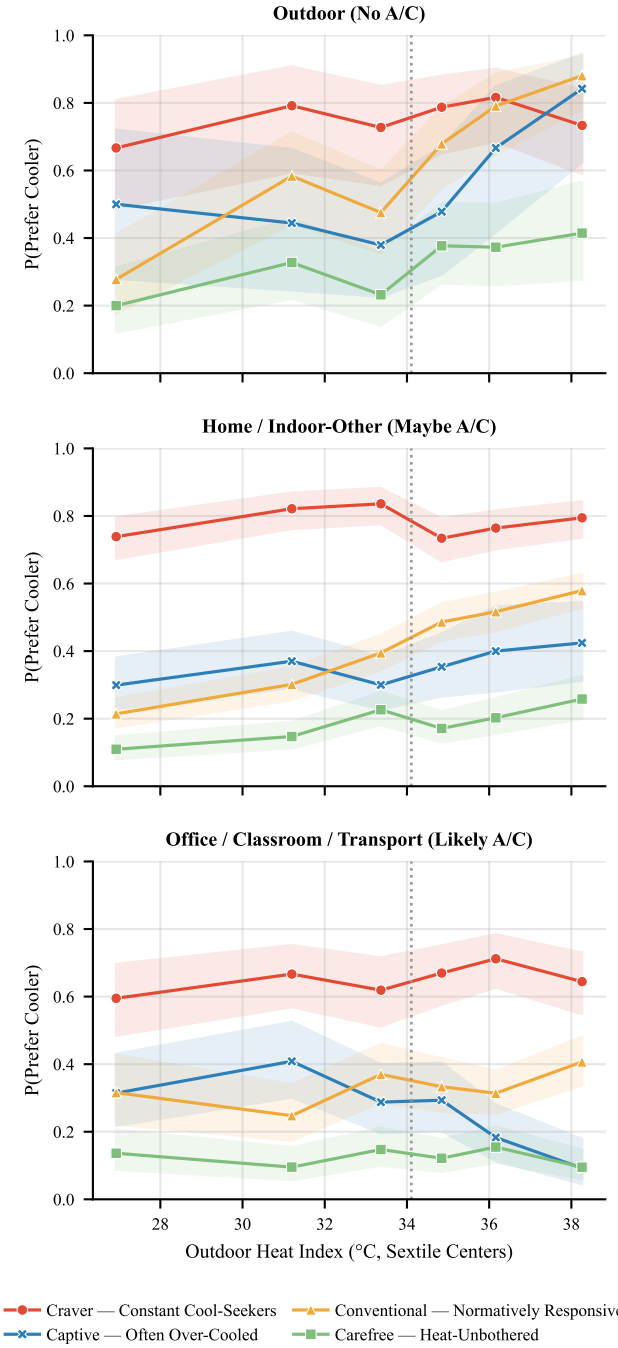


Figure 10: How each archetype responds to rising outdoor heat, shown as the probability of preferring cooler conditions across heat-index sextiles, stratified by probable control context from lowest (top) to highest (bottom) air-conditioning likelihood, with Wilson 95 percent intervals and the cohort median heat index.

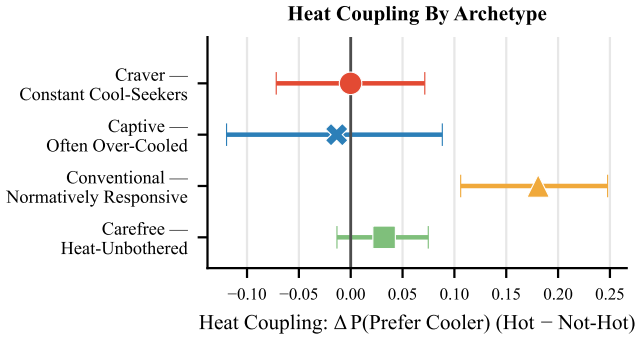


Figure 11: How tightly each archetype's cooling demand tracks heat, shown as the difference in participants' prefer-cooler probability between hot and not-hot observations, with 95% bootstrap confidence intervals.

4. Discussion

4.1. Role of behavioral archetypes in thermal comfort research for the built environment

As outlined in the introduction, there is a long history of characterizing how the human body reacts to its thermal environment, from heat-balance models of comfort to the physiology of thermoregulation and cutaneous blood flow [1, 26, 2]. This work describes, in principle, the conditions under which a person should feel comfortable, yet real-world thermal behavior is consistently more complex than the models predict [7, 8]. The link between what a person's body senses and what that person then decides to do, whether to seek, purchase, or operate cooling, is shaped by expectation, habit, and circumstance as much as by physiology [9, 34, 35]. The archetypes found here show that gap in a tropical setting. The *Conventional* behave largely as physiological theory would predict, tracking outdoor heat in the pattern expected from thermoregulation [26]. The *Captive*, like most groups, ask for cooling less often in likely air-conditioned spaces, but they stand out for being too cold, an over-cooling consistent with critiques of low tropical set points and of comfort standards that drive over-cooling in summer [20, 48]. The *Craver* and *Carefree* are harder to reconcile with the physical environment alone: the *Craver* want cooling whatever the heat or setting, and the *Carefree* rarely do, even outdoors in above-median heat, with neither group's cooling demand tracking outdoor heat. Preferences of this kind are unlikely to be explained by heat balance and instead point to the psychological, cultural, and economic factors on the right side of Figure 1, from expectation and the pleasantness of a stimulus given internal state [9, 32] to the social conventions and material cultures that normalize particular indoor climates [17, 36, 50]. Isolating which of these factors drive the *Craver* and the *Carefree* will require field-based, longitudinal studies that combine behavioral measurement with the qualitative and mixed methods better suited to motivation and meaning [35, 37].

4.2. Implications for the design and operations of buildings

These findings bear on how buildings are designed and operated, since the conventional approach is largely one-size-fits-

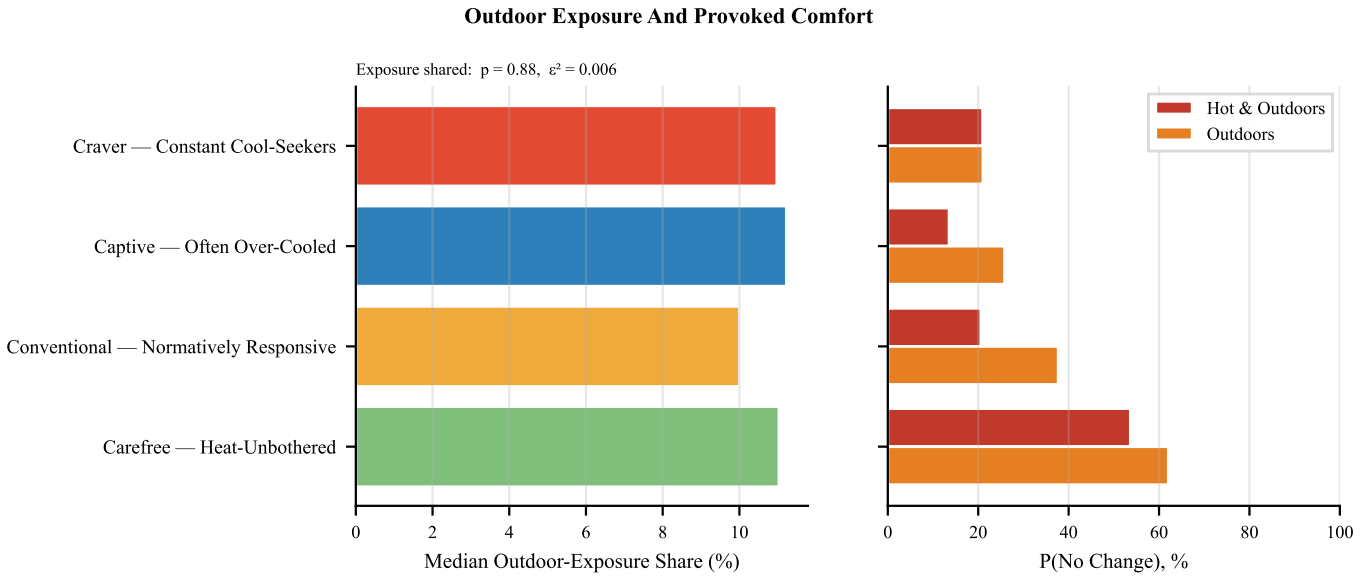


Figure 12: Where the archetypes spend their time and how they cope when heat is unavoidable: outdoor-exposure share (left) and the probability of reporting No Change when outdoors and when outdoors in above-median heat (right).

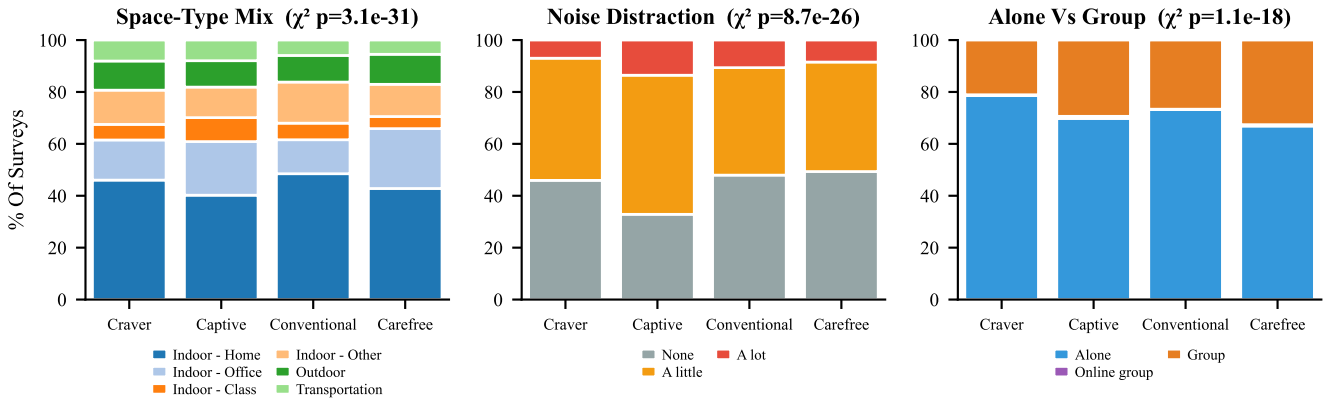


Figure 13: Survey characteristics held out from the clustering, offering an independent check on the archetypes: space type, noise distraction, and social context (left to right).

all, providing a single set point intended to satisfy an average occupant [7]. A more holistic approach would combine several routes to comfort so that occupants can choose what works for them, whether through adaptive operation, personally controlled air movement, or personal comfort systems [8, 20, 12]. The residential context is where this matters most, since households themselves decide whether to install cooling and how to run it, decisions that respond to income, price, and habit as well as to physiology [44, 41, 43]. Air conditioning remains the most effective way to cool indoor space in a warming climate [52], heat is among the most commonly reported causes of disrupted sleep in hot, humid cities [62], and the *Conventional* group shows that much cooling demand is a real response to heat. Cooling to low set points is not required for productivity, however, as office work performance has been found to hold steady between 23 and 29 °C [21]. The concern is the *Craver* pattern of heat-independent cooling demand, which, if it is common across a rapidly air-conditioning tropical belt, could carry large consequences for energy use and carbon emissions [44, 45, 46, 47]. Addressing it will require working with social scientists and behavioral economists to understand cooling as a practice, and to find a balance between the habitual overuse of air conditioning and its habitual avoidance [17, 37, 18]. Greater thoughtfulness about when and where cooling is installed, together with a broader palette of technologies such as radiant systems, elevated air movement, adaptive clothing, and personal cooling, could deliver comfort to more of these archetypes at a fraction of the energy cost [5, 20, 23].

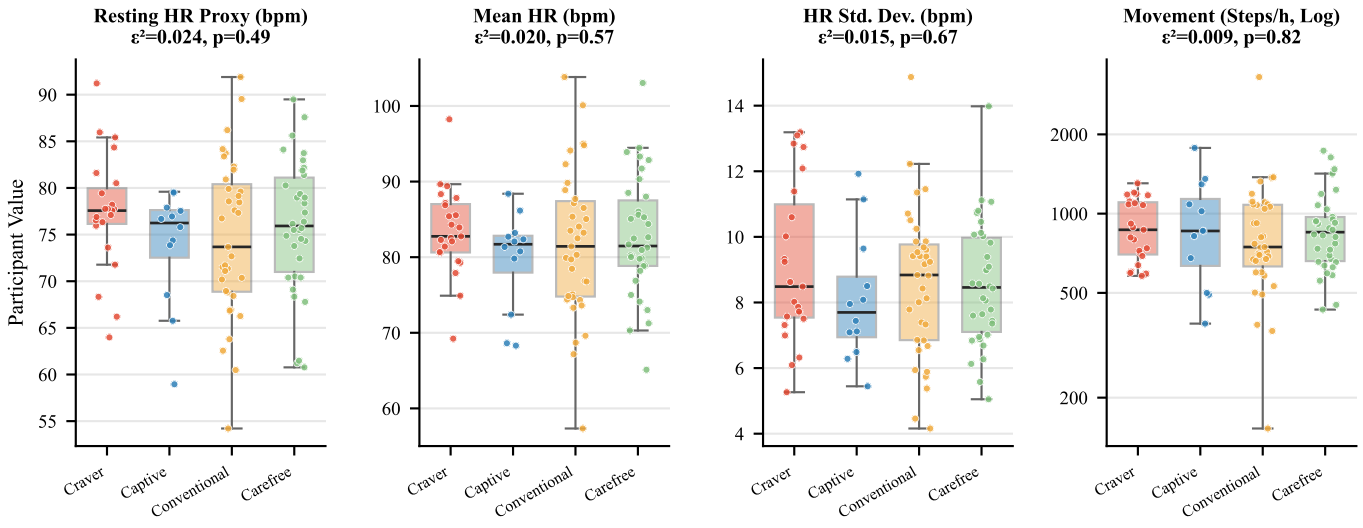


Figure 14: Wearable physiology offers no shortcut to the archetypes: participant-level summaries during the hour before each survey, comprising resting heart-rate proxy, mean heart rate, heart-rate standard deviation, and movement (left to right). None differs significantly across archetypes (Kruskal-Wallis epsilon-squared < 0.03).

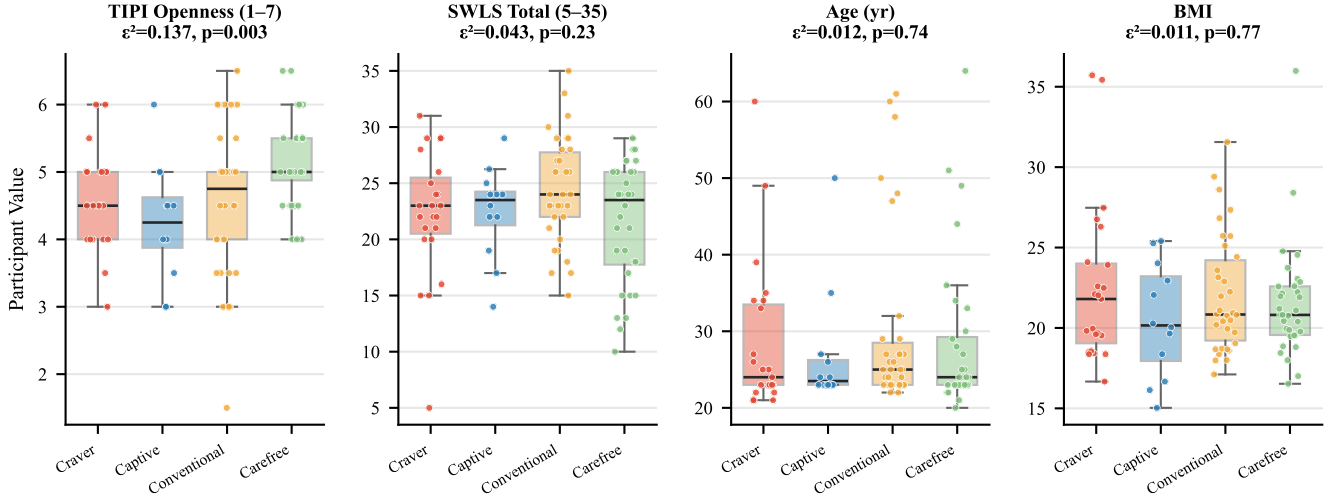


Figure 15: One stable trait stands out among the onboarding measures (n = 101): openness, life satisfaction, age, and body mass index (left to right). Openness runs higher among the *Carefree* archetype (Kruskal-Wallis epsilon-squared = 0.137, p = 0.003), while the other measures overlap across groups.

4.3. Limitations

This work has several limitations. The environmental-control contexts are defined a priori from the likelihood that a space is air-conditioned, so statements about likely air-conditioned settings describe association with space type and must not be read as causal claims about air-conditioning operation, which was not measured directly. The cohort is drawn from a single tropical city and is young and largely self-selected, which limits generalization to other climates, age groups, and cultures [16, 15]. Because survey observations are nested within participants and were not experimentally assigned, every comparison is descriptive and associational rather than causal. Outdoor weather is only a proxy for each participant's immediate microclimate, so indoor conditions and radiant effects are only partially captured [5]. The location and control-context assignments rest on self-reported space type and on the assumption that Singapore offices, classrooms, and transport are usually air-conditioned, which is plausible in aggregate but cannot be confirmed for individual responses. The response distribution is imbalanced, with prefer-warmer votes especially rare, so the smaller archetypes such as the *Captive* are estimated with wider uncertainty. Finally, the archetype names are descriptive labels applied after clustering, and membership is soft near the boundaries, so the four groups should be read as an interpretable discretization of a continuous behavioral space rather than as fixed types.

4.4. Future work

The most important next step is replication in other climates and cohorts, with independent validation of whether a similar set of archetypes emerges beyond a young tropical sample [16]. Pairing the wearable micro-surveys with direct indoor environmental measurements, including air and radiant temperature and personal exposure, would remove the reliance on outdoor weather as a proxy and sharpen the environmental-control contrasts [5]. Following participants over longer periods would show whether archetype membership is stable or whether preferences drift as expectations and thermal history change [9, 8]. Moving beyond association will require designs that can evaluate causal control interventions, for example just-in-time adaptive interventions that nudge cooling behavior and measure the response [56]. Explaining why the *Craver* and the *Carefree* diverge will ultimately depend on integrating the psychological, cultural, and economic disciplines represented on the right side of Figure 1, through mixed-methods collaboration with the behavioral and social sciences [17, 37, 18].

5. Conclusions

This study used a longitudinal smartwatch micro-survey deployment in Singapore, comprising 9,358 geolocated thermal-preference responses from 104 eligible participants, to derive a behavioral typology of thermal preference. Clustering three person-level features (the rate of preferring cooler conditions, the rate of preferring warmer conditions, and the sensitivity of cooling demand to outdoor heat) yielded four archetypes: the *Craver*, the *Captive*, the *Conventional*, and the *Carefree*. Only the *Conventional* showed cooling demand that rises with outdoor heat, and every group except the *Craver* asked for cooling less often in spaces likely to be air-conditioned. The *Craver* sought cooling whatever the heat or setting, the *Carefree* seldom did even in hot outdoor conditions, and the *Captive* were distinguished by frequently feeling too cold. These patterns are stable under reasonable perturbations of the clustering and are not explained by wearable physiology or by most stable personal traits; only personality openness shows a moderate association. Thermal preference in this warm climate therefore appears to be only partly a physiological response, shaped also by behavior and, plausibly, by expectation and habit, which argues for comfort strategies and cooling policies that let people choose among low-carbon ways of staying comfortable.

CRediT authorship contribution statement

Clayton Miller: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Resources, Writing – original draft, Visualization, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The manuscript source, analysis documentation, and publication figures for this paper are available at `https://github.com/buds-lab/cooling-cravers-to-carefree-journal-paper`.
The analysis draws on two public data sources linked by pseudonymous participant identifiers. The privacy-protected Cozie Singapore survey-level dataset, which contains the smartwatch micro-survey responses, matched outdoor weather, and wearable-health aggregates used to construct the thermal-preference archetypes, is available at `https://github.com/buds-lab/cozie-singapore-journal`.
The onboarding dataset used to examine participant demographics, personality traits, life satisfaction, sensory sensitivity, and acclimatization is available in the Make Yourself Comfortable repository at `https://github.com/buds-lab/make-yourself-comfortable-jitai-journal-paper`.
Both source repositories provide their own data documentation and privacy notes, and no private participant information is included in this manuscript repository.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used Runcell AI (Version 0.1.25), an AI coding and writing assistant built on the GPT-5.6 Terra (OpenAI) and Claude Opus 4.8 (Anthropic) large language models, to help organize and format the LaTeX manuscript, arrange figure and table placement, and copy-edit text written by the author. The tools were not used to generate or analyze the study data, nor to create or alter any figure, its underlying values, or the reported results. All AI-assisted output was reviewed, edited, and verified by the author, who takes full responsibility for the originality, accuracy, integrity, citations, and final content of this manuscript.

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